

StoreRx

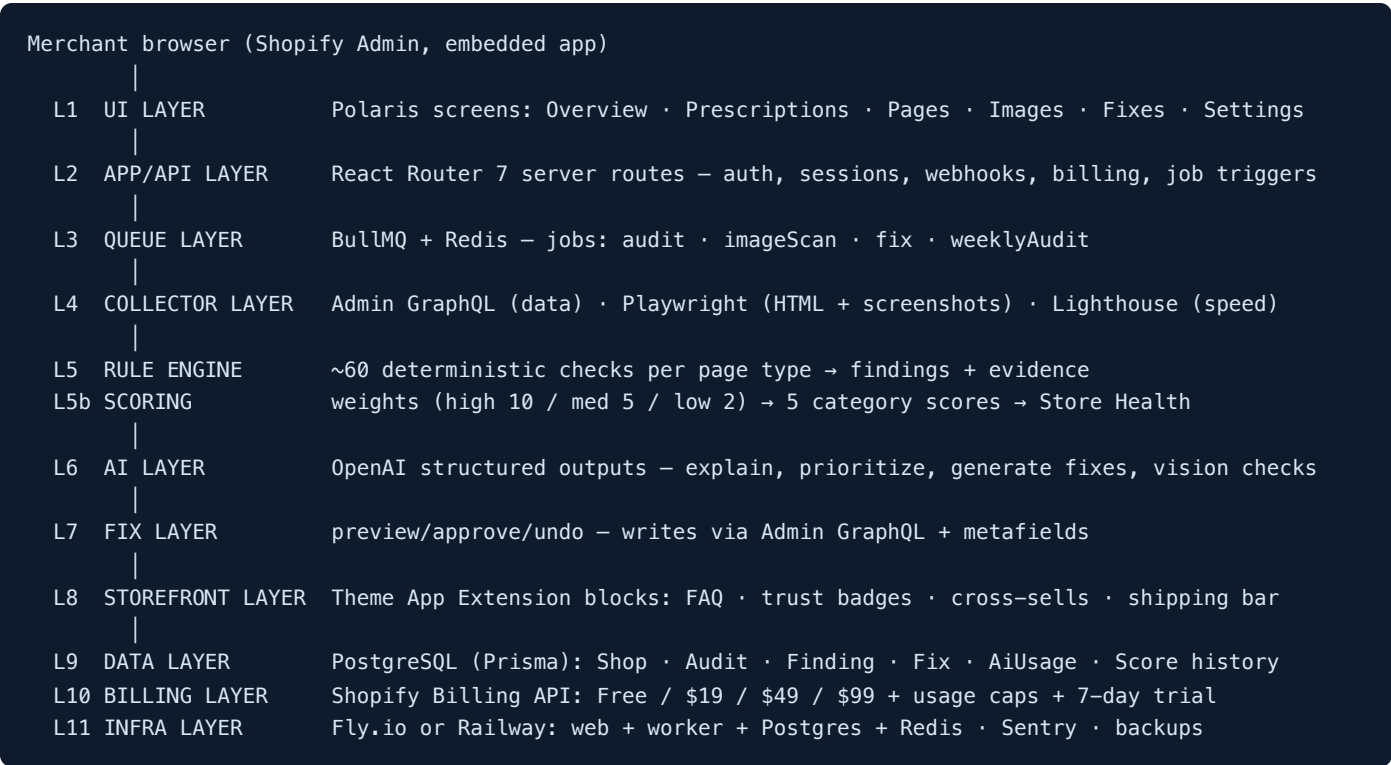
Diagnose. Optimize. Convert. — AI Conversion Doctor for Shopify · Full build plan · September 2026

What it does: A merchant installs the app. StoreRx scans the store page by page (Homepage → Collection → Product → Cart → Checkout), gives scores for Conversion, UX, Performance, SEO and Product Pages, shows a prioritized list of problems ("prescriptions"), and can fix many of them with AI — always with merchant preview and approval, never silently.

1. Golden rules

- **Code finds problems. AI explains and fixes.** Every check is a deterministic rule; the AI never "guesses" whether an issue exists.
- **AI never changes the store without approval.** Every fix: preview → approve → apply → undo (30 days).
- **No invented numbers.** Impact is High / Medium / Low — never "+12% conversion".
- **All AI calls use structured JSON output** through one wrapper function (easy to swap OpenAI ↔ Claude later).
- **Heavy work runs in background workers** — never inside a web request.
- **Storefront changes only via Theme App Extension** — never edit the merchant's theme code.

2. All layers of the system



Layer	Technology	Responsibility
L1 UI	Polaris, App Bridge	Embedded dashboard the merchant sees inside Shopify admin.
L2 App/API	React Router 7 + @shopify/shopify-app-react-router	OAuth, sessions, GDPR webhooks, billing routes, starts jobs. No heavy work here.
L3 Queue	BullMQ + Redis	Runs audits (1–3 min) and fixes in background; retries; progress updates to UI.
L4 Collectors	Admin GraphQL, Playwright, Lighthouse	Gathers facts: products, media, top sellers, page HTML, mobile+desktop screenshots, speed metrics.
L5 Rules + Scoring	TypeScript, unit-tested	Deterministic checks (e.g. "reviews above fold?", "image > 500 KB?"). Produces findings with evidence, then scores.
L6 AI	OpenAI SDK (gpt-4.1-mini / gpt-4.1)	Explains findings in merchant language, ranks priority, writes descriptions / SEO / FAQ / alt text, judges image quality (vision).
L7 Fix	Admin GraphQL mutations	Applies approved fixes; keeps "before" snapshot; undo for 30 days.
L8 Storefront	Theme App Extension (Liquid, <10 KB JS)	Blocks merchant adds in theme editor: FAQ, badges, cross-sells, shipping bar.
L9 Data	PostgreSQL + Prisma	Shops, audits, findings, fixes, AI token usage, score history for trend graphs.
L10 Billing	Shopify Billing API	4 plans, trial, per-plan AI usage caps to protect margins.
L11 Infra	Fly.io / Railway, Sentry	Hosting, scaling, monitoring, logs, backups.

3. What gets checked (summary)

Page	Main checks
Homepage	Hero CTA above fold (mobile), announcement bar, featured products, trust elements, popup count, navigation, footer links
Collection	Filters/sort, price on cards, quick add-to-cart, thin/empty collections, image consistency, description
Product	Reviews above fold, sticky CTA, price/stock/delivery near CTA, description length, FAQ/size guide, trust badges, cross-sells, out-of-stock variants, SEO title/meta, structured data
Cart	Drawer, free-shipping bar, upsells, trust badges, express checkout buttons
Checkout	(via Admin API) Shop Pay / Apple Pay enabled, guest checkout, shipping & payment options
Images	Size >500 KB, dimensions, missing alt text, mixed ratios, <3 images, duplicates + AI vision: clutter, watermark, blur, off-center
Performance	Lighthouse LCP/CLS/INP per page + which installed apps add how many KB of scripts

Audit samples pages (home, 3 collections, 5 top-selling products, cart) — full image catalog is a separate job on paid plans.

4. AI fixes ("Fix with AI")

Fix	Model	Applied to	Plan
Product description	gpt-4.1	product.descriptionHtml	\$19+
SEO title + meta description	gpt-4.1-mini	product.seo	\$19+
Image alt text	gpt-4.1-mini (vision)	media alt	Free (20) / unlimited \$19+
FAQ section	gpt-4.1-mini	metafield → theme block	\$19+
Trust badges / CTA copy / cross-sells / shipping bar	gpt-4.1-mini	theme extension blocks	\$49+
Image compression / background removal	sharp / image API	re-upload after approval	\$19+ / \$49+

AI cost per merchant: audit $\approx$ \$0.02 · one fix $\approx$ \$0.005–0.03 · full vision scan of 200 products $\approx$ \$1–2. Under \$1/month for a typical \$19 merchant — margins are safe.

5. Production deployment

Item	Choice	Notes
Hosting	Fly.io or Railway	2 processes: web (app) + worker (audits/fixes). Worker needs 2 GB RAM for Playwright + Lighthouse.
Database	Managed PostgreSQL	Daily backups on. Prisma migrations in deploy step.
Redis	Managed (Upstash / Fly Redis)	Queues + audit progress.
Secrets	Platform env vars	SHOPIFY_API_KEY/SECRET, DATABASE_URL, REDIS_URL, OPENAI_API_KEY, SENTRY_DSN. Never in git.
CI/CD	GitHub Actions	test → build → deploy on push to main; <code>shopify app deploy</code> for extension versions.
Monitoring	Sentry + platform logs	Alert on failed jobs, webhook errors, OpenAI error rate.
Scaling	Queue concurrency	Lighthouse concurrency 1–2 per worker; add workers as installs grow.
Costs (start)	$\approx$ \$25–50/month	Hosting + DB + Redis + Sentry free tier + OpenAI usage.

App Store launch checklist

- GDPR webhooks (customers/data_request, customers/redact, shop/redact) — implemented and tested
- Billing works for all 4 plans + trial; uninstall cleans data within 48 h
- Privacy policy (mentions OpenAI processing, no training on data), support email
- Listing: name check (StoreRx trademark/domain!), icon, 6 screenshots, demo video
- Review wait: 1–3 weeks, expect 1–2 rounds of feedback

6. Pricing

Free	\$19/mo Starter	\$49/mo Growth	\$99/mo Pro
1 audit/month, 5 AI recommendations, 20 alt texts	Weekly audits, unlimited recommendations, product copy fixes, image compression	Unlimited audits, all AI fixes, full vision scan, background removal, conversion analytics	Everything + multi-store, advanced analytics, priority support

7. Timeline — full-time plan (6 h/day, 6 days/week)

Stage 1 — MVP build: ~1 month (26 working days, ≈ 120 hours)

Days	Work	Hours
Day 1–3	Setup: auth, dev store link, database models (Shop, Audit, Finding, Fix, AiUsage), billing 4 plans + trial	15
Day 4–8	Collectors: Admin GraphQL, Playwright crawl + screenshots, Lighthouse, BullMQ queue	25
Day 9–12	Rule engine: 12 product rules + image checks + scoring + fixture tests	18
Day 13–15	AI layer: generate() wrapper, structured outputs, prompts (explain, description, SEO, alt text), token logging	12
Day 16–19	Fix workflow: preview / approve / apply / undo, bulk alt text	18
Day 20–24	Dashboard UI (Polaris): Overview, Prescriptions, Fixes, Settings	22
Day 25–26	Weekly scheduled audits + email summary	8
Total	Working MVP on a dev store	≈ 118 h

Stage 2 — Production hardening: 15–20 days (≈ 90–120 hours)

Days	Work	Hours
Day 1–3	Hosting: Fly.io/Railway web + worker, managed Postgres + Redis, secrets, deploys	15
Day 4–6	CI/CD (GitHub Actions), Sentry monitoring, logging, backups	15
Day 7–11	Real-theme edge cases: password pages, redirects, broken themes, slow stores, retries	28
Day 12–14	GDPR webhooks, uninstall cleanup, billing edge cases (upgrade/downgrade/decline)	16
Day 15–18	Performance + cost hardening: queue concurrency, AI usage caps, rate limits, security pass	20
Day 19–20	Homepage + collection + cart rules (expand beyond product pages)	12

Stage 3 — Testing & launch: 6–10 days (≈ 40–60 hours)

Days	Work	Hours
Day 1–4	Beta test on 3–5 real stores, fix bugs found, polish copy and empty states	25
Day 5–6	App Store listing: name check, icon, 6 screenshots, demo video, privacy policy, support email	12
Day 7–10	Submit for review + answer reviewer feedback (review wait itself is 1–3 weeks — keep improving during it)	15

Total: ~2 months full-time from first line of code to App Store submission ($\approx$ 250–300 hours). The Shopify review wait (1–3 weeks) runs after Stage 3 — use that time for the vision image scan, theme blocks, and trend graphs (+60–80 h post-launch features).

Repo: [~/storerx](#) — official Shopify React Router 7 template, scaffolded and committed. Spec: [docs/FEATURES.md](#).